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Genome-Wide Association Study of Intraocular Pressure in Population-Based Cohorts in Japan: The Tohoku Medical Megabank Organization Eye Study

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Purpose: This study was conducted to elucidate the distribution and determinants of ocular biometric parameters and to assess the association between intraocular pressure (IOP) and single nucleotide polymorphisms (SNPs) in the Japanese population—based genome cohort studies.

Design: Cross-sectional analysis involving genome-wide association studies (GWASs).

Participants: In total, 22 150 participants aged >18 years from the population cohort (Community-Based Cohort [CommCohort]) and 11 302 participants from the Birth and Three-Generation (BirThree) Cohort of the Tohoku Medical Megabank Organization Eye Study were examined.

Methods: Participant underwent interviews, ophthalmic and physiological examinations, laboratory tests, and microarray analyses. Genome-wide association studies were conducted in the CommCohort (discovery stage) and the BirThree Cohort (replication stage), followed by a meta-analysis. Associations of SNPs and IOP were evaluated using a genome-wide significance threshold (5×10^{-8}).

Main Outcome Measures: Association of SNPs with IOP and distributions of IOP by sex and age.

Results: In the discovery stage, the mean IOP of the right and left eye was 13.95 and 14.02 mmHg, respectively. In the replication stage, the corresponding values were 14.32 and 14.27 mmHg, respectively. A significant age-related reduction in IOP was observed in both stages ($P < 0.001$). Genome-wide association studies identified 573 and 2 genome-wide significant SNPs in the discovery and replication stages, respectively. Meta-analysis revealed 1601 significant SNPs across 21 loci on 11 chromosomes (Chrs). Of these loci, 17 were previously known to be associated with IOP or glaucoma, while four—septin-8 (*SEPT8*; Chr5), aldehyde dehydrogenase 2 (*ALDH2*; Chr12), collagen type VI alpha 2 chain (*COL6A2*; Chr21), and Wnt family member 7B (*WNT7B*; Chr22)—were newly identified.

Conclusions: This large-scale GWAS in a Japanese population identified 21 loci associated with IOP, including 4 novel loci. The findings highlight both genetic similarities and population-specific variations in SNPs influencing IOP and provide valuable insights to enhance eye health care, including glaucoma management.

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Elevated intraocular pressure (IOP) represents the most critical risk factor and a robust predictor for the development of glaucoma, a condition characterized by glaucomatous optic nerve degeneration and progressive visual field loss. Primary open-angle glaucoma (POAG) is the leading cause of irreversible blindness worldwide.[1,2] The only reliable and modifiable risk factor for the onset and progression of glaucoma is elevated IOP,[3,4] and its reduction remains the sole therapeutic approach to mitigating glaucomatous progression, even in cases of normal-tension glaucoma.[5,6] Despite elevated IOP being a risk factor for the most common form of glaucoma, POAG, more than half of glaucoma cases remain undiagnosed until irreversible optic nerve damage has occurred.[2,7] In addition to elevated IOP,

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other major risk factors for POAG include advancing age, sex, ethnicity, and a positive family history.[7,8]

The endophenotype of glaucoma like cup area, disc area, and IOP exhibits high heritability.[8] One approach to disentangle the genetic underpinnings of the complex etiology of glaucoma is to focus on IOP, which has demonstrated genetic correlations with POAG. Heritability estimates for IOP range from 0.29 to 0.36 based on sibling studies.[9,10] Higher estimates of heritability have been reported in other cohorts, including 0.42 in a multicenter study of Caucasian POAG families[11] and even higher values of 0.68 and 0.40 in studies of monozygotic and dizygotic twins, respectively.[12]

Although the onset of glaucoma has been extensively investigated through epidemiological approaches, yielding numerous significant findings regarding IOP alterations and glaucoma development, the underlying biological mechanisms driving these changes remain inadequately understood. Consequently, it is increasingly vital to elucidate the genetic factors associated with IOP and glaucoma onset to deepen our understanding of the disease's etiology. Notably, early genome-wide association studies (GWASs) have collectively identified over 100 loci linked to IOP.[13–21] Furthermore, genetic investigations have demonstrated that the genetic risk factors for elevated IOP and POAG are partially shared. For example, the transmembrane and coiled-coil domains 1 (*TMCO1*) and growth arrest specific 7 (GAS7) genes have been implicated in both increased IOP and the pathogenesis of POAG.[21–23]

The Tohoku Medical Megabank (TMM) has been established since 2013. Tohoku University and Iwate Medical University have been collaborating closely to conduct the TMM project, and the project operating at the Tohoku University is acronymized as Tohoku Medical Megabank Organization (ToMMo). Tohoku Medical Megabank manages a comprehensive biobank founded on 2 uniquely structured population cohorts.[24] These cohorts have recruited participants from the general population through 2 strategically designed frameworks: the community cohort and the birth-and-three-generation cohort. The TMM project conducts comprehensive physiological assessments for participants during the baseline survey and subsequent follow-up surveys, approximately every 5 years.[24,25] These assessments include specialized ophthalmic evaluations, collectively known as the ToMMo Eye Study. The ToMMo Eye Study examines IOP, axial length (AL), and refractive errors and performs fundoscopy.[26] In parallel, TMM undertakes microarray-based genomic analyses, complemented by genotype imputation utilizing both Japonica DNA microarrays—designed specifically for the Japanese population,[27,28] and whole-genome sequence reference panels.[29] To date, approximately 33 000 participants have undergone ophthalmic evaluations during the baseline survey, with microarray analyses completed for all these individuals.

Investigating the genetic determinants of IOP offers valuable insights into the genetic architecture of POAG and normal-tension glaucoma. Understanding the genes influencing IOP may unveil novel pathways for glaucoma treatment. In this study, we conducted GWAS utilizing IOP measurements and microarray data from participants of the ToMMo Eye Study. This research leveraged the strategically designed 2 cohorts of the TMM project. Here, we present the findings of the GWAS for IOP and explore the genetic relationships between single nucleotide polymorphisms (SNPs) and IOP.

Methods

Cohort Design and Study Population

Tohoku Medical Megabank is conducting a population-based genome cohort study and constructing the integrated biobank, which includes ToMMo Study (Tohoku University) and IMM study (Iwate Medical University).[24] In TMM, 2 prospective cohort studies have been established strategically: the Community-Based Cohort (CommCohort) Study, which recruited 84 073 participants,[25] and the Birth and Three-Generation (BirThree) Cohort Study, which recruited 73 529 participants.[30] More than 157 000 individuals participated voluntarily, and approximately 33 000 participants in ToMMo cohort studies received additional detailed baseline examinations for health conditions including ophthalmic examinations at 7 community support (assessment) centers,[24] which is referred to as the ToMMo Eye Study. Approximately 65% of residents who we asked were enrolled with informed consent in the ToMMo CommCohort Study. The BirThree Cohort Study recruited pregnant participants at the obstetric hospital, and the enrollment rate was similar.

In this project, we conducted a set of ophthalmic examinations for the participants in all 7 community support centers, including assessments of IOP and AL and examination of the fundus. We also obtained the following sets of information from the participants: (1) answers to a wide range of questionnaires regarding family history, residential area, education, and various environmental factors; (2) results of physiological examinations; and (3) results of multiomics analyses for participant biospecimens. We excluded participants with a history of glaucoma, based on the information from the questionnaire (discovery stage: 914 participants; validation stage: 49 participants) because there were some cases where it was unclear whether they had undergone laser surgery, surgery, or only medication. Information on ocular disease was also obtained from questionnaires. We excluded pregnant women from this IOP analysis because of the decrease in IOP during pregnancy.[31,32]

After removing the subjects who met the exclusion criteria, such as inconsistent with gender, identical twins, extreme outlier individuals, we utilized the CommCohort Study with 22 150 genotyped subjects with IOP measurements for the discovery-stage GWAS on IOP. We also used the BirThree Cohort Study with 11 302 genotyped subjects with IOP measurements for the validation-stage GWAS on IOP. For the meta-analysis stage, we analyzed 33 452 genotyped subjects of both cohort studies.

Our genome medical research coordinators informed eligible individuals the aims and protocols of the TMM study and obtained their informed consent. At the baseline examination, we explained the protocols of the cohort studies, biobanking, and the research methods for genomic analyses and omics analyses. Institutional review board approval was obtained from the Ethics Committee of the TMM Organization (2012-4-565, 2013-4-103, 2020-4-155, and 2020-4-156). This study was conducted in accordance with the Declaration of Helsinki, the Ethical Guidelines for Human Genome or Gene Analysis Research, and other appropriate guidelines.

Eye Examination Procedures

Comprehensive ophthalmic evaluations encompassed the measurement of IOP and AL. Intraocular pressure was measured 3 times for each eye, beginning with the right eye, and the average value was calculated using a fully automated tonometer, TX-20P (Canon). Axial length was assessed using the OA-1000 (Tomey), which uses partial coherence interferometry to capture signals from the tear film and retinal pigment epithelium.

Genotyping, Imputation, and Quality Control

To prepare direct and imputed genotype datasets, 33 833 samples from the CommCohort Study and the BirThree Cohort Study were genotyped with an Affymetrix Axiom Japonica Array (v2). For quality control, we excluded plates with an average call rate <0.95 and removed samples with a DishQC metric <0.82 or a step1 call rate <0.97 before batch genotyping. We excluded samples with unusually high identical by descent values compared with most of the other samples. We applied an SNP quality control step to each batch to exclude variants with a Hardy−Weinberg equilibrium test P value $< 1.00 \times 10^{-5}$, a minor allele frequency <0.01, or a missing rate >0.01 by using PLINK (v1.60b6.24 and v2.00a).[33] We merged the imputed genotype datasets using QCTOOL (v2.0.4) (https://www.well.ox.ac.uk/~gav/qctool/).

We phased the genotype data of each batch using SHAPEIT2 (v2. r837).[34] Genotype imputation was performed for the phased genotype data of each batch with IMPUTE2 (ver. 2.3.2)[35] using a phased reference panel of 3552 Japanese individuals from cohort studies of the TMM project.[29] We then merged the imputed genotype datasets using QCTOOL (v2.0.4) (https://www.well.ox.ac.uk/~gav/qctool/). Ultimately, we obtained an imputed genotype dataset in Oxford BGEN format and a direct genotype dataset in PLINK BED format.

We applied an SNP quality control step to the imputed and direct genotype datasets for the discovery and validation stages to exclude variants with Hardy−Weinberg equilibrium test P values $<1.00 \times 10^{-5}$, minor allele frequency values <0.01, or missing rates >0.05. The number of remaining variants in each dataset for GWAS was 9 310 609 in the discovery stage and 9 302 721 in the validation stage. Then, we extracted 22 437 individuals who were from the CommCohort Study along with their IOP phenotypes and covariate values for the discovery stage, and we extracted 11 396 individuals who were from the BirThree Cohort Study along with their IOP phenotypes and covariate values for the validation stage. After exclusion of subjects who showed gender or sex inconsistency, identical twins, extreme outlier individuals as detected by principal component analysis, and subjects with a history of ocular disease as determined by questionnaires, we selected 22 150 subjects for the discovery stage and 11 302 subjects for the validation stage.

GWASs and Statistical Analysis

We used BOLT-LMM v2.3.6[36] for linear mixed model analysis to test the additive genetic effects of SNPs on IOP. In our GWAS analysis, biases from the population stratification as well as familial and cryptic relatedness were controlled by considering the genetic correlation matrix in the linear mixed model.[37] To reduce the skewness and kurtosis of the distribution of IOP and ages, Box-Cox transformation was applied to these data with an R package (car ver. 4.1.2). We adjusted for age and sex as covariates in the IOP analysis. For the IOP correction, we used conversion formula from a previous Japanese central corneal thickness (CCT) study[38] and coefficient of the IOP correction formula and used corrected IOP (mmHg) = IOP reading −0.012 (reading CCT − mean CCT) (CCT). The mean CCT values for the ultimately extracted populations are 544.03 μm for the CommCohort and 542.82 μm for the BirThree Cohort.

In GWAS for each phenotype, we input the direct genotype dataset and the imputed genotype dataset into BOLT-LMM. We merged the results for the direct genotype dataset and the results for the imputed genotype dataset by overwriting the latter results with the former results for variants existing in both datasets. We examined associated SNPs in the discovery GWAS in the CommCohort utilizing the BirThree Cohort as the validation cohort. Genetic heritability was estimated by using linkage disequilibrium score regression.[39] For meta-analysis genome-wide association scans, we used METAL software (ver. 2011-03-25).[40] Regional association plots for the target regions were generated using LocusZoom (ver. 1.4).[41] We visualized the recombination rates estimated with the BirThree Cohort study[42] on the LocusZoom plots. To obtain the effect sizes on the millimeters of mercury scale for the SNPs that were significant in the meta-analysis, we merged the discovery stage and validation stage populations and performed association analysis by using plink2. Additionally, we performed further association analyses using a dominant model with plink2.

Results

Study Populations and Baseline Characteristics of the ToMMo Eye Study

During recruitment into the 2 TMM cohorts, participants underwent comprehensive baseline assessments. As part of this baseline study, a series of ophthalmic evaluations were conducted at community support centers, including measurements of IOP, AL, and fundus imaging. Additionally, the following datasets were collected from participants:

i. Responses to extensive questionnaires covering family history, residential information, educational background, and various environmental factors.

ii. Results from more than 10 physiological examinations.

iii. Multiomics analysis data. These datasets formed the foundation of the ToMMo Eye Study. Here, we present the findings of association analyses between

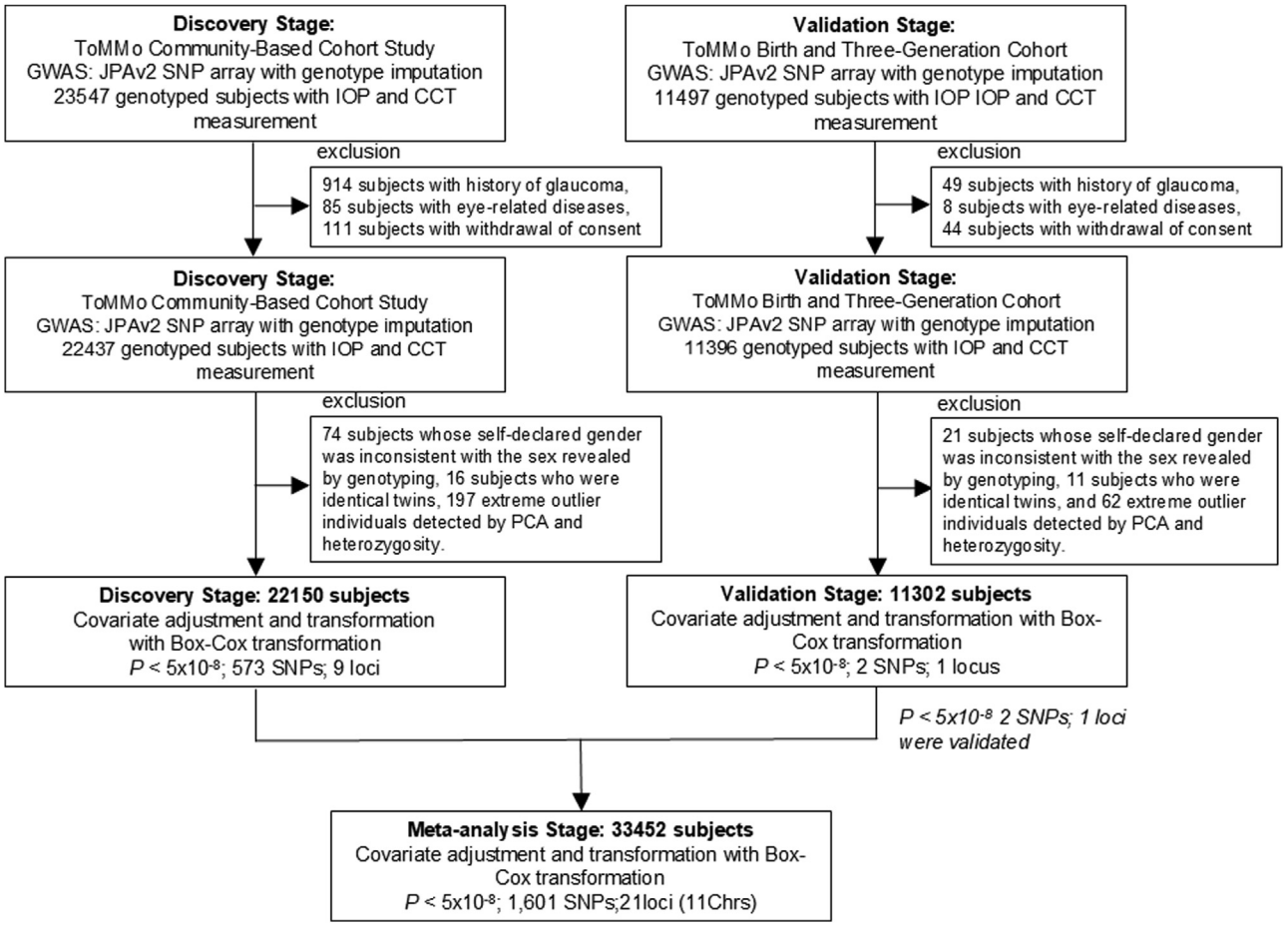


Figure 1. Genome-wide association study design for investigating genetic correlates of IOP across 2 cohorts. The inclusion criteria for the 2 population cohorts and the overall scheme of GWAS for the discovery and validation stages are presented. The discovery-stage GWAS utilized the CommCohort Study, comprising 22 150 genotyped participants with IOP measurements. The validation-stage GWAS was conducted using the BirThree Cohort Study, which included 11 302 genotyped participants with IOP measurements, after applying the exclusion criteria. Associations between SNPs and IOP were assessed using a genome-wide significance threshold ($P < 5 \times 10^{-8}$) in each stage and in a combined meta-analysis involving a total of 33 452 participants. BirThree = Birth and Three-Generation; CCT = central corneal thickness; CommCohort = Community-Based Cohort; GWAS = genome-wide association study; IOP = intraocular pressure; PCA = principal component analysis; SNP = single nucleotide polymorphism.

IOP, genomic data, questionnaire responses, and physiological examination results.

We conducted a GWAS at the discovery stage using imputed microarray genotype data and IOP measurements from 22 437 participants of the CommCohort Study. For the validation stage, a similar analysis was performed on data from 11 396 participants of the BirThree Cohort (Fig 1). For the discovery-stage GWAS, the following exclusions were applied: 74 participants due to inconsistencies between self-reported gender and genetically determined sex, 16 participants with identical twins, and 197 extreme outliers identified through principal component analysis. After these exclusions, 22 150 individuals were included in the

Table 1. Baseline Characteristics of Participants in the ToMMo Eye Study

	Community-Based Cohort	Birth and Three-Generation Cohort
Number, total	22 150	11 302
Number, male (%)	6806 (30.7)	4180 (37.0)
Mean age, year	61.76 (SD: 12.40)	43.73 (SD: 13.48)
Intraocular pressure (mmHg) right	13.95 [13.91–13.99]	14.32 [14.27–14.38]
Intraocular pressure (mmHg) left	14.02 [13.98–14.06]	14.27 [14.21–14.33]

SD = standard deviation; ToMMo = Tohoku Medical Megabank Organization.
95% CI (confidence interval) in square brackets.

discovery-stage GWAS. Similarly, for the validation-stage GWAS, the exclusions included: 21 participants with gender or sex inconsistencies, 11 participants with identical twins, and 62 extreme outliers identified by principal component analysis. This resulted in a final sample size of 11 302 individuals for the validation-stage GWAS.

As presented in Table 1, IOP measurements were collected from 22 150 participants in the TMM CommCohort Study, comprising 6806 men and 15 344 women, with a mean age of 61.76 ± 12.40 years. Similarly, IOP data were obtained from 11 302 participants in the BirThree Cohort, including 4180 men and 7122 women, with a mean age of 43.73 ± 13.48 years. The predominance of female participants in both studies reflects the demographic characteristics typical of prospective cohort studies.

The IOP assessments in these 2 cohort studies were performed at identical facilities, namely the community support centers. However, a critical distinction between the studies warrants attention for accurate interpretation of the findings. Specifically, the age distribution of participants varied significantly between the 2 cohorts (Fig 2). Participants in the discovery study were predominantly aged 60 to 70 years, whereas those in the validation study were mainly in their 30s and 60s. The overall IOP distribution across the population exhibited a moderate rightward skew (Fig 3).

IOP Variations by Laterality, Sex, and Age Groups

Although no significant differences in IOP values were observed between the discovery and validation stages, we hypothesized that IOP measurements might exhibit substantial variation based on sex, age, and laterality (right vs. left eyes). Utilizing the IOP data collected from a total of 33 452 participants, we explored this hypothesis in detail. The distributions of IOP values by age, sex, and eye laterality are presented in Table 2.

An examination of the data revealed a significant age-associated decline in IOP values among both men and women ($P < 0.001$) during the discovery stage (Fig 4, left panel). This age-related reduction in IOP was consistently observed in the validation stage data (Fig 4, right panel). Given that the validation cohort includes participants from the BirThree Study, encompassing a substantial proportion of younger individuals, our findings regarding the age-related decline in IOP among Japanese individuals appear robust. These results stand in stark contrast to studies on European and African American populations, where IOP values typically increase with age.[43] Additionally, we observed that IOP values were generally lower in women compared to men in the discovery stage data, a pattern that was similarly replicated in the validation stage (Table 2). This study thus highlights the presence of

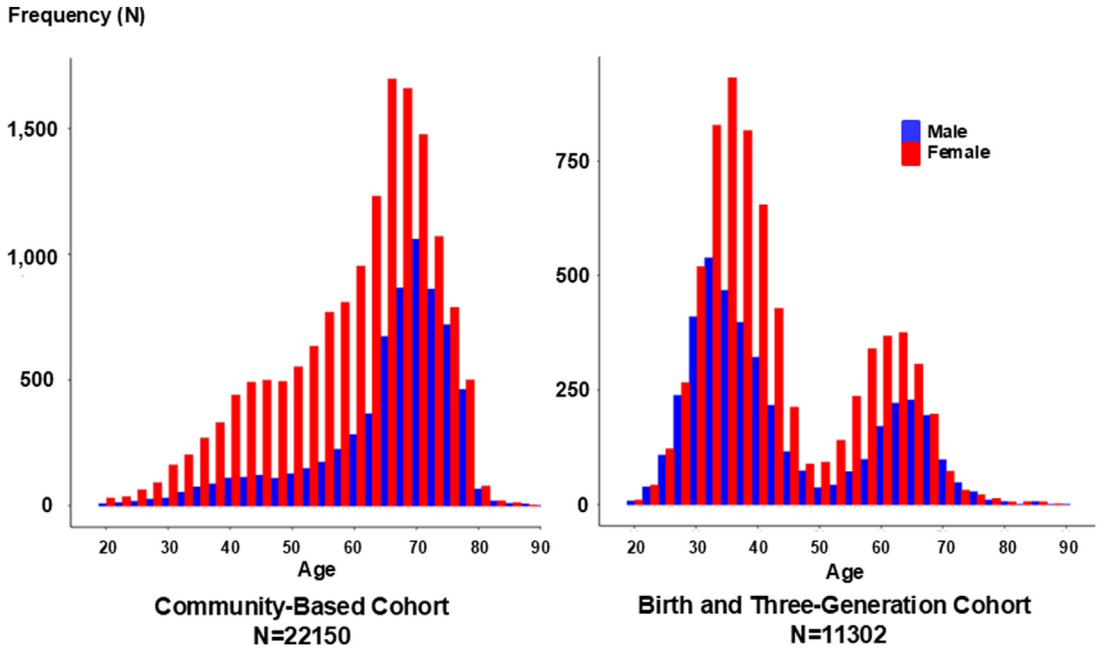


Figure 2. Age distribution of the study population. Intraocular pressure examinations were conducted at 7 community support centers. For the discovery stage, a total of 22 150 participants were recruited (mean age 61.76 ± 12.40 years; 6806 men and 15 344 women). For the validation stage, 11 302 participants were included (mean age 43.73 ± 13.48 years; 4180 men and 7122 women). In the discovery stage, the majority of participants were in their 60s to 70s, whereas the validation stage exhibited 2 age peaks, primarily in the 30s and 60s.

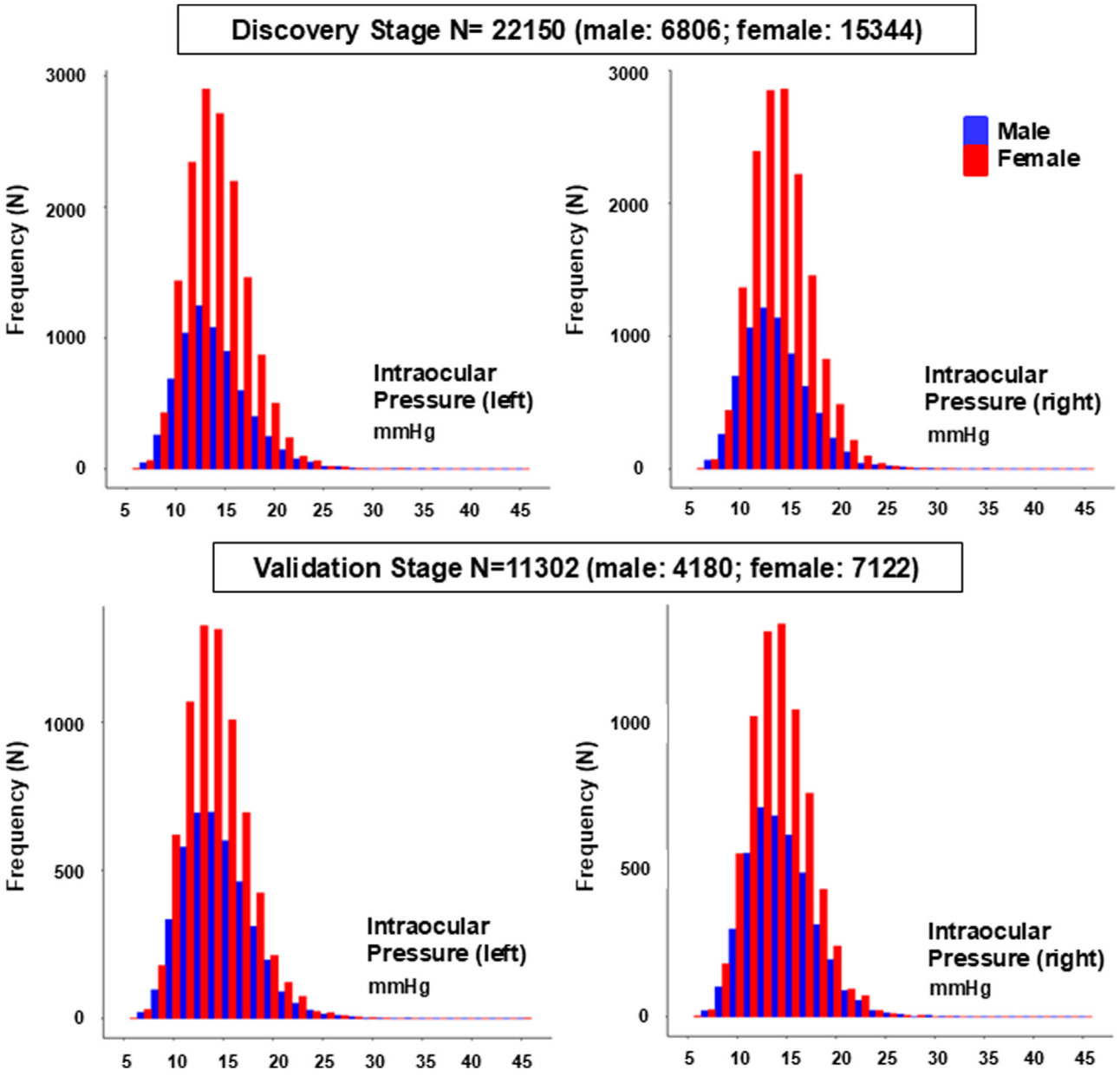


Figure 3. Intraocular pressure distribution across age groups. The mean IOP was 13.95 mmHg (95% CI: 13.91–13.99) in the right eye and 14.32 mmHg (95% CI: 14.27–14.38) in the left eye during the discovery stage. In the validation stage, the mean IOP was 14.02 mmHg (95% CI: 13.98–14.06) in the right eye and 14.27 mmHg (95% CI: 14.21–14.33) in the left eye. Notably, IOP in both the right and left eyes exhibited significant age-related variations among both female and male participants. CI = confidence interval; IOP = intraocular pressure.

ethnic differences in age-related IOP changes, although the underlying mechanisms driving these disparities remain to be elucidated.

An additional noteworthy finding in this study is the asymmetry in IOP values between the right and left eye (Fig 5). The most pronounced differences were observed among younger participants (Table 2). In the CommCohort, the right eye exhibited significantly higher IOP than the left eye in participants in their 30s ($P = 0.035$). Similarly, in the Birth and BirThree Cohort, the right eye showed higher IOP values than the left in participants under 30 and in their 30s ($P = 0.009$ and $P = 0.007$, respectively). Notably, in participants under 50 years of age, the IOP of the right eye was higher than that of the left eye in both the discovery and validation stages. This relationship reversed in participants >50 years of age, with the left eye showing higher IOP values in both men and women (not statistically significant), except for women in the validation stage, where the right eye IOP remained higher in those >60 years of age. Additionally, we observed that variations in IOP values were generally greater in the right eye compared to the left eye (Fig 5).

However, when aggregated across all age groups, the differences in IOP between the right and left eyes were less

Table 2. Age Distribution of Intraocular Pressure in the Tohoku Medical Megabank Organization Eye Study

Intraocular Pressure (CommCohort: Discovery Cohort)									
Variable	*Whole Population (N = 22 150)*							*Total*	P *Value**
Age (yrs)	<30	30–39	40–49	50–59	60–69	70–79	≥80		
N	340	1352	2392	3556	8110	6268	132	22 150	
IOP right eye (R) (mmHg)									
mean	15.41	14.86	14.52	14.11	13.88	13.48	13.02	13.95	<0.001
95% CI	[15.07, 15.76]	[14.69, 15.03]	[14.39, 14.64]	[14.01, 14.21]	[13.81, 13.94]	[13.41, 13.56]	[12.46, 13.58]	[13.91, 13.99]	
IOP (L) (mmHg)									
Mean	14.97	14.61	14.46	14.24	13.96	13.63	13.34	14.02	<0.001
95% CI	[14.63, 15.32]	[14.44, 14.78]	[14.34, 14.59]	[14.14, 14.34]	[13.90, 14.03]	[13.55, 13.71]	[12.66, 14.01]	[13.98, 14.06]	
Men (N = 6806)									
N	67	276	454	708	2388	2850	63	6806	
IOP (R) (mmHg)									
Mean	15.82	15.32	15.09	14.49	13.94	13.46	12.87	13.94	<0.001
95% CI	[14.92, 16.72]	[14.91, 15.73]	[14.78, 15.39]	[14.25, 14.73]	[13.81, 14.07]	[13.34, 13.58]	[12.04, 13.71]	[13.86, 14.02]	
IOP (L) (mmHg)									
Mean	15.53	15.08	15.00	14.71	14.02	13.67	12.90	14.06	<0.001
95% CI	[14.68, 16.38]	[14.67, 15.48]	[14.68, 15.32]	[14.46, 14.96]	[13.89, 14.15]	[13.54, 13.8]	[11.99, 13.81]	[13.97, 14.14]	
Women (N = 15 344)									
N	273	1076	1938	2848	5722	3418	69	15 344	
IOP (R) (mmHg)									
Mean	15.32	14.75	14.38	14.02	13.85	13.50	13.15	13.96	<0.001
95% CI	[14.94, 15.69]	[14.56, 14.93]	[14.25, 14.52]	[13.91, 14.13]	[13.78, 13.93]	[13.4, 13.6]	[12.38, 13.93]	[13.91, 14.00]	
IOP (L) (mmHg)									
Mean	14.84	14.49	14.34	14.12	13.94	13.59	13.74	14.00	<0.001
95% CI	[14.46, 15.21]	[14.3, 14.67]	[14.2, 14.47]	[14.01, 14.23]	[13.86, 14.02]	[13.49, 13.7]	[12.74, 14.74]	[13.95, 14.05]	
Intraocular Pressure (BirThree Cohort: Validation Cohort)									
Variable	*Whole Population (N = 11 302)*							*Total*	P *Value**
Age (yrs)	<30	30–39	40–49	50–59	60–69	70–79	≥80		
N	1161	4998	1722	1181	1965	250	25	11 302	
IOP (R) (mmHg)									
Mean	14.71	14.43	14.29	14.16	14.13	13.28	12.08	14.32	<0.001
95% CI	[14.53, 14.89]	[14.34, 14.51]	[14.14, 14.44]	[13.98, 14.34]	[13.99, 14.27]	[12.88, 13.68]	[10.97, 13.2]	[14.27, 14.38]	
IOP (L) (mmHg)									
Mean	14.40	14.29	14.36	14.20	14.26	13.40	12.20	14.27	<0.001
95% CI	[14.21, 14.58]	[14.21, 14.38]	[14.2, 14.52]	[14.02, 14.38]	[14.11, 14.41]	[13.02, 13.77]	[11.02, 13.38]	[14.21, 14.33]	
Men (N = 4180)									
N	543	1808	591	292	798	139	9	4180	
IOP (R) (mmHg)									
Mean	15.03	14.70	14.56	14.34	14.25	13.21	12.61	14.56	<0.001
95% CI	[14.75, 15.3]	[14.55, 14.85]	[14.28, 14.84]	[13.97, 14.71]	[14.02, 14.48]	[12.63, 13.79]	[10.02, 15.2]	[14.46, 14.66]	
IOP (L) (mmHg)									
Mean	14.73	14.60	14.70	14.53	14.36	13.36	12.50	14.53	<0.001
95% CI	[14.46, 15]	[14.45, 14.75]	[14.41, 14.99]	[14.13, 14.93]	[14.12, 14.61]	[12.82, 13.89]	[10.17, 14.83]	[14.43, 14.64]	
Women (N = 7122)									
N	618	3190	1131	889	1167	111	16	7122	

(*Continued*)

Table 2. (Continued.)

Intraocular Pressure (BirThree Cohort: Validation Cohort)									
Variable	*Whole Population (N = 11 302)*							*Total*	P *Value**
IOP (R) (mmHg)									
Mean	14.43	14.27	14.15	14.10	14.04	13.38	11.79	14.19	<0.001
95% CI	[14.19, 14.67]	[14.17, 14.38]	[13.97, 14.32]	[13.9, 14.3]	[13.86, 14.21]	[12.82, 13.93]	[10.55, 13.03]	[14.12, 14.26]	
IOP (L) (mmHg)									
Mean	14.10	14.12	14.18	14.09	14.18	13.45	12.03	14.12	0.037
95% CI	[13.86, 14.35]	[14.01, 14.23]	[14, 14.37]	[13.9, 14.29]	[13.99, 14.38]	[12.93, 13.96]	[10.5, 13.56]	[14.05, 14.19]	

CI = confidence interval; IOP = intraocular pressure.
*Kruskal–Wallis rank sum test.

pronounced. In the discovery stage, the mean IOP of the right eye in men was 13.94 mmHg (95% confidence interval [CI]: 13.86–14.02), while in women, it was 13.96 mmHg (95% CI: 13.91–14.00). For the left eye, the mean IOP in men was 14.06 mmHg (95% CI: 13.97–14.14), and, in women, it was 14.00 mmHg (95% CI: 13.95–14.05) (Table 2).

Background Characteristics Associated with IOP

We examined the relationships between IOP values and various background characteristics of the participants, with the findings summarized in Table 3 and 4. The analysis revealed that CCT, serum glucose levels, total cholesterol, platelet count, lung function (measured as percent-predicted forced expiratory volume in the first second), and both systolic and diastolic blood pressures were strongly associated with variations in IOP values (Fig 6). Notably, the correlation between IOP and CCT was particularly pronounced, with Pearson correlation coefficients (r) reaching absolute values of approximately 0.3 in the discovery cohort (Table 3 and 4). Similar trends were observed in the validation cohort, although the correlation coefficients between IOP and CCT were higher in the validation cohort compared to the discovery cohort (Table 3 and 4).

Except CCT, diastolic and systolic blood pressures emerged as the most significant determinants of IOP among the background factors analyzed. Notably, diastolic blood pressure demonstrated stronger correlations with IOP compared to systolic blood pressure in our study. The Pearson correlation coefficients (r) between diastolic blood pressure and IOP in the right and left eyes were 0.128 and 0.135,

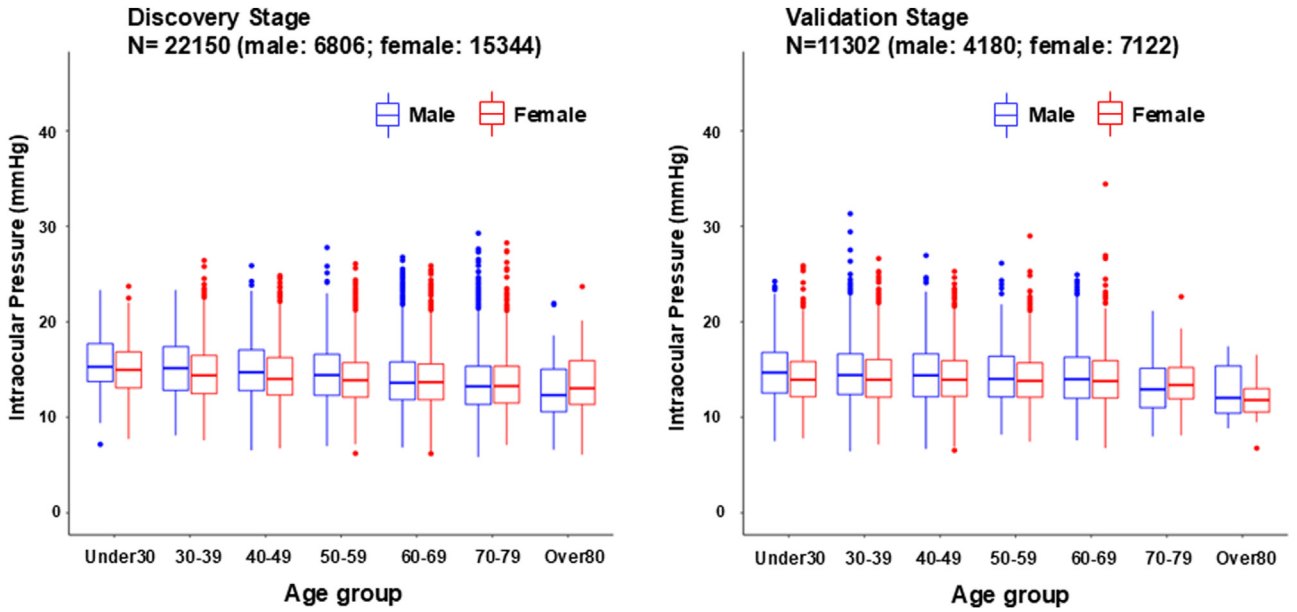


Figure 4. Age-related decrease in IOP values showed a significant decline with advancing age in both men and women ($P < 0.001$) during the discovery stage (left panel). This age-associated reduction in IOP was consistently reproduced in the validation-stage data (right panel). IOP = intraocular pressure.

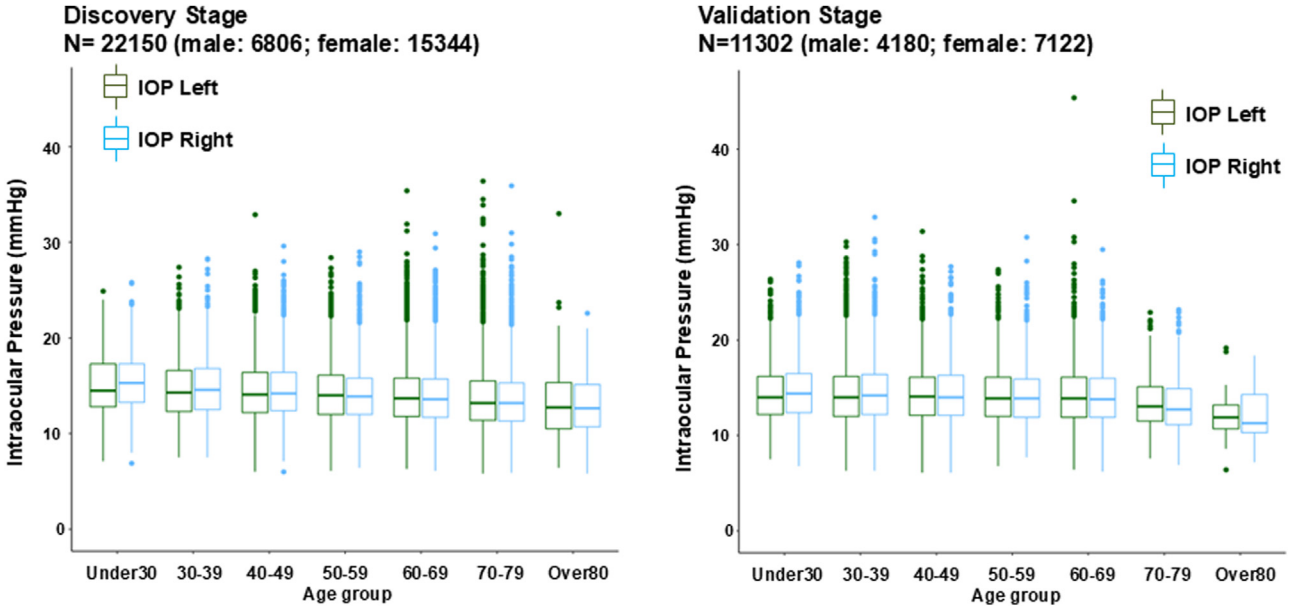


Figure 5. Age-dependent right−left differences in IOP. The most pronounced differences in IOP between the right and left eyes were observed among younger participants. During the discovery stage, the right eye exhibited significantly higher IOP than the left eye in individuals in their 30s ($P < 0.035$, left panel). Similarly, in the validation stage, the right eye showed significantly higher IOP compared to the left eye in participants under 30 ($P = 0.009$) and in their 30s ($P = 0.007$, right panel). This trend persisted in participants under 50 years of age in both stages. However, the relationship reversed in older age groups, with the left eye demonstrating higher IOP values in elderly participants. IOP = intraocular pressure.

respectively, in the discovery cohort, and 0.168 and 0.190, respectively, in the validation cohort (Table 3 and 4). For systolic blood pressure, the corresponding coefficients were 0.093 and 0.109 in the discovery cohort and 0.136 and 0.156 in the validation cohort. These findings indicate that the correlations were consistently stronger in the validation cohort compared to the discovery cohort. Furthermore, these results diverge markedly from those of the UK Biobank Eye Study, where systolic blood pressure was identified as the most critical background determinant of IOP.[43]

A particularly intriguing finding is the significant negative correlation between IOP and age, as illustrated in Figure 4. This observation contrasts sharply with the results of the UK Biobank study,[43] which reported a positive correlation between IOP and age. While the latter finding aligns with the well-documented increase in glaucoma incidence with advancing age,[44] our observation from the 2 cohort studies may offer valuable insights into the

Table 3. Associations between Intraocular Pressure and Background Characteristics Community-Based Cohort (Discovery Cohort)

	Age	IOP (R)	IOP (L)
Age	-		
IOP (R)	−0.142	-	
IOP (L)	−0.104	**0.774**	-
CCT (R)	0.005	**0.308**	**0.293**
CCT (L)	−0.003	**0.298**	**0.298**
Glucose	**0.232**	0.027	0.039
Total cholesterol	0.061	0.058	0.062
Platelet number	**−0.230**	0.078	0.062
FEV1%	**−0.308**	0.050	0.037
Height	**−0.204**	0.018	0.018
Systolic blood pressure	**0.412**	0.093	0.109
Diastolic blood pressure	0.063	0.128	0.135

CCT = central corneal thickness; FEV1% = percent-predicted forced expiratory volume in the first second; IOP = intraocular pressure.
n = 19 758 (male = 6059, female = 13 699), Each column shows Pearson correlation coefficients (r), absolute values of 0.2 or more are in bold, *P* value adjustment method: Bonferroni $P < 0.003$, correlation coefficient: $r > |0.024|$ is significant.

Table 4. Associations between IOP and Background Characteristics Birth and Three-Generation Cohort (Validation Cohort)

	Age	IOP (R)	IOP (L)
Age	-		
IOP (R)	−−0.078	-	
IOP (L)	−0.035	**0.773**	-
CCT (R)	0.029	**0.345**	**0.326**
CCT (L)	0.027	**0.336**	**0.332**
Glucose	**0.260**	0.034	0.046
Total cCholesterol	**0.299**	0.067	0.080
Platelet number	**−0.203**	0.060	0.050
FEV1%	**−0.418**	0.046	0.025
Height	**−0.247**	0.021	0.020
Systolic blood pressure	**0.477**	0.136	0.156
Diastolic blood pressure	**0.286**	0.168	0.190

CCT = central corneal thickness; FEV1% = percent-predicted forced expiratory volume in the first second; IOP = intraocular pressure.
n = 9774 (male = 3730, female = 6044), Each column shows Pearson correlation coefficients (r), absolute values of 0.2 or more are in bold, *P* value adjustment method: Bonferroni $P < 0.003$, correlation coefficient: $r > |0.032|$ is significant.

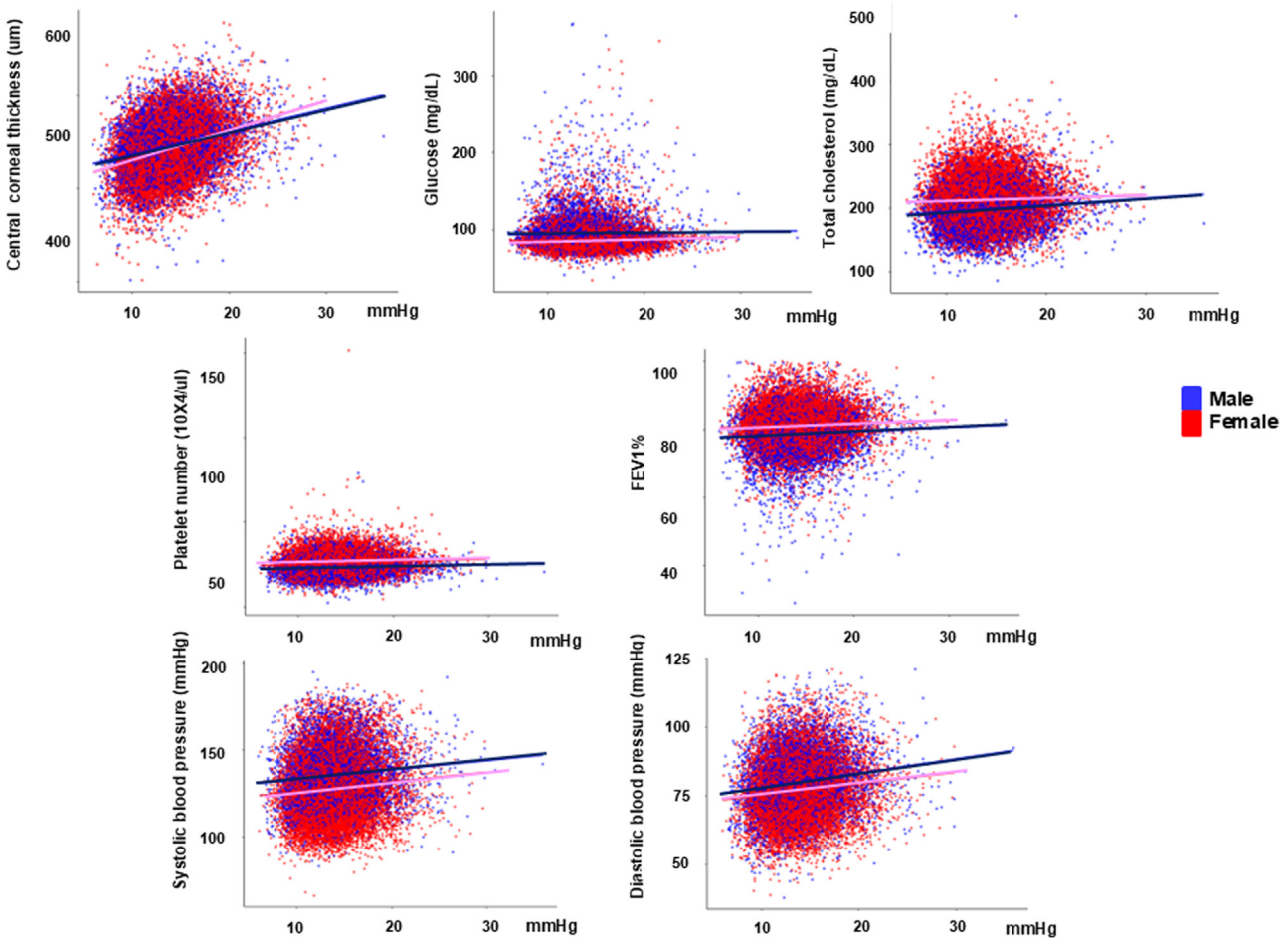


Figure 6. Background characteristics associated with IOP. The associations between IOP values and participants' background characteristics were analyzed. Significant correlations were observed between IOP and several factors, including central corneal thickness, serum glucose levels, total cholesterol levels, platelet count, lung function (percent-predicted forced expiratory volume in the first second), as well as systolic and diastolic blood pressures (data from both discovery stages are presented). IOP = intraocular pressure.

pathogenesis of glaucoma in the Japanese population and highlight potential ethnic differences in the mechanisms underlying POAG pathogenesis.

GWAS Identified 8 Loci in the ToMMo Eye Study

We performed a GWAS on in IOP using microarray data from postquality control samples in the CommCohort Study (discovery stage) and the BirThree Cohort Study (validation stage). Following rigorous quality control procedures, the analyses incorporated 574 162 directly genotyped SNPs and 8 736 447 imputed SNPs in the discovery stage, alongside 577 888 directly genotyped SNPs and 8 724 833 imputed SNPs in the validation stage. Quantile–quantile plots demonstrated evidence of mild genomic inflation, with the linkage disequilibrium score intercept values of 0.9941 and 1.0146 for the discovery and validation stages, respectively (Fig S1, available at www.ophthalmologyscience.org).

We identified 573 SNPs that surpassed the genome-wide significance threshold of $P < 5 \times 10^{-8}$ in the discovery-stage screening (Fig 7). These SNPs were grouped into 8 distinct genomic regions. Among these 8 loci, 7 have previously been reported as being associated with IOP or glaucoma.[15–18,20,21,45–47] Additionally, we identified a novel locus: *ALDH2* (aldehyde dehydrogenase 2) on chromosome (Chr) 12 (Table 5). Of the 8 loci, the 4 most significant associations were found in the TLC domain containing 5 (*TLCD5*)~Rho Guanine Nucleotide Exchange Factor 12 (*ARHGEF12*) locus (Online Mendelian Inheritance in Man [OMIM]: 604763; rs56118776; $P = 2.4 \times 10^{-13}$) on Chr11q23.3, *ANAPC1* (anaphase promoting complex subunit 1) locus (OMIM: 608473; rs34739497; $P = 2.9 \times 10^{-13}$) on Chr2q13, *GAS7* locus (OMIM: 603127; rs9913911; $P = 4.0 \times 10^{-13}$) on Chr17p13.1.,[18,21,47] and *FNDC3B* (fibronectin type III domain-containing protein 3B) locus (OMIM: 611909; rs10589611; $P = 8.6 \times 10^{-12}$) on Chr 3q26.31.[14,18,21,48] We also conducted similar GWAS using the validation-stage data and identified 2 SNPs within the *FNDC3B* locus that surpassed the genome-wide significance threshold of $P < 5 \times 10^{-8}$ (Fig 7). This locus corresponds to the top hit identified in the discovery-stage GWAS.

Although the SNPs reaching genome-wide significance in the validation stage were limited to a single locus,

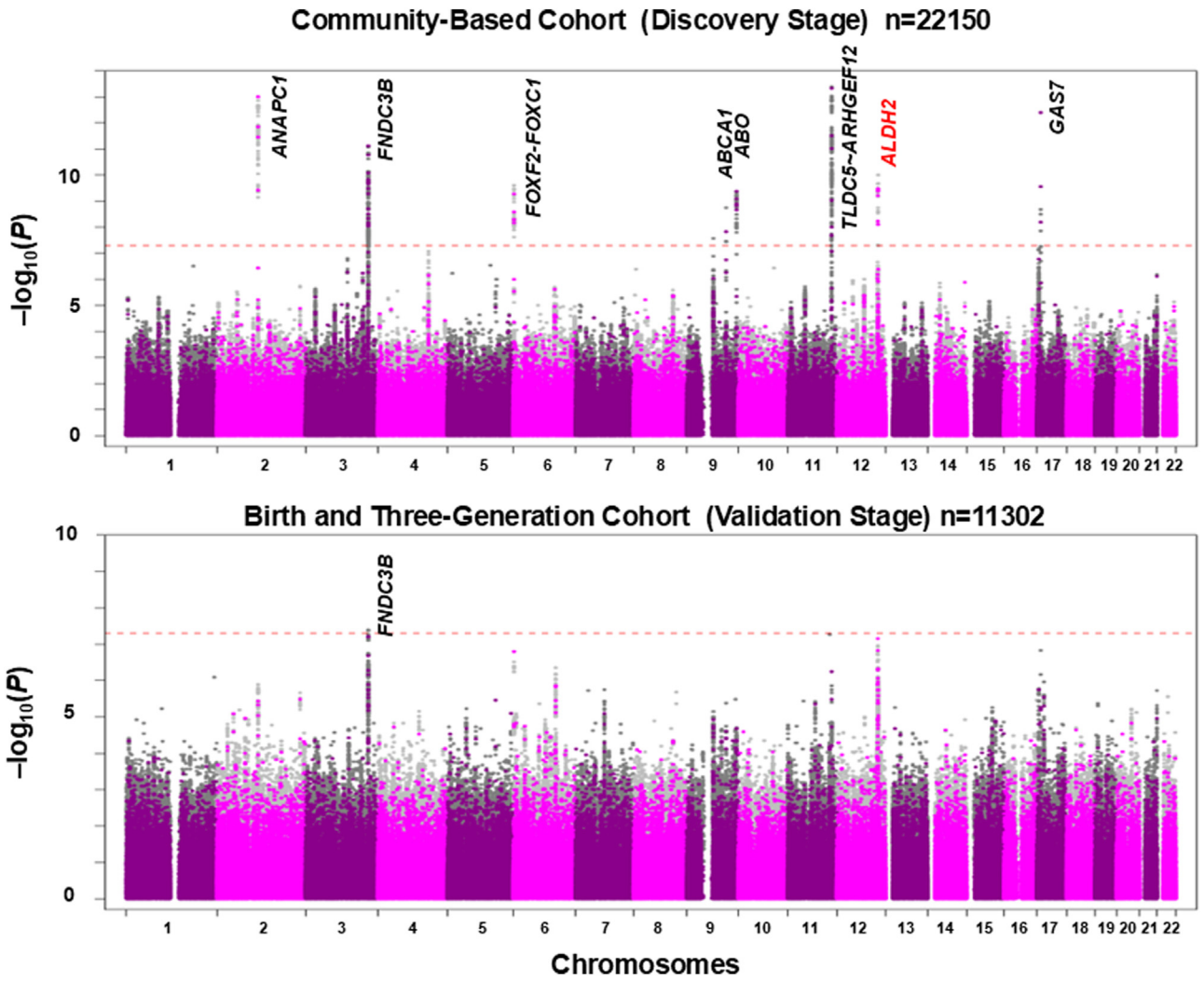


Figure 7. Genome-wide association study of IOP in the discovery and validation stage. The results of the GWAS for IOP conducted in the discovery and validation stages are presented. Data for both directly genotyped and imputed SNPs are visualized in the Manhattan plot. The y-axis depicts $-\log_{10}$(P) values for associations with IOP, while the x-axis represents chromosomes and base-pair positions based on human genome build 37. The horizontal red dotted line denotes the genome-wide significance threshold ($P < 5.0 \times 10^{-8}$). Previously reported genes (loci) are labeled in black, while newly identified genes (locus) from this study are highlighted in red. *ABO* = alpha 1-3-N-acetylgalactosaminyltransferase and alpha 1-3-galactosyltransferase; *ALDH2* = aldehyde dehydrogenase 2; *ANAPC1* = anaphase promoting complex subunit 1; *FNDC3B* = fibronectin type III domain-containing protein 3B; *FOXF2* = forkhead box F2; *GAS7* = growth arrest specific 7; GWAS = genome-wide association study; IOP = intraocular pressure; SNP = single nucleotide polymorphism.

considerably fewer than those identified in the discovery stage, it is noteworthy that the SNPs detected during the discovery stage exhibited relatively high statistical significance in the validation stage. While the results demonstrate a moderate correlation between the findings of the 2 stages, the statistical power of the GWAS in the validation cohort was insufficient to fully replicate all 8 loci identified in the discovery stage. Nonetheless, these findings strongly suggest that the significant SNPs identified in the discovery stage were reliably corroborated in the validation stage.

Meta-Analysis of 2 GWAS for the ToMMo Eye Study

Given the insufficient statistical power of the validation cohort to achieve significant replications, we conducted a meta-analysis combining the discovery-stage and validation-stage GWAS results. Using the METAL software, the meta-analysis identified 21 loci across 11 Chrs (Fig 8). The basic characteristics of these loci are summarized in Table 5. For reference, the meta-analysis combining the discovery-stage and validation-stage GWAS results using the METAL software and the meta-analysis data using plink2 are shown in Table S1 (available at www.ophthalmologyscience.org). Among these 21 loci, four—septin-8 (*SEPT8)* (Chr 5), *ALDH2* (Chr 12), collagen type VI alpha 2 chain (*COL6A2*) (Chr 21), and Wnt family member 7B (*WNT7B*) (Chr 22)—were newly identified in this study, while the remaining 17 loci had previously been reported as associated with IOP or glaucoma. Our study successfully reproduced and validated these loci.[15–18,20,21,45–47] Among the 21 loci,

Table 5. Loci and Lead SNPs Associated with Intraocular Pressure: A Total of 21 Loci

Chr	Lead SNP*	Position†	Nearest Gene	RA	EA	IMP	EA Freq 54KJPN	Discovery Stage			Validation Stage			Meta-Analysis		
								B‡	SE	P	B‡	SE	P	B§	SE	P
2	rs34739497	112,625 267	*ANAPC1*	CA	C	1	0.1817	0.0232	0.0032	**2.9E-13**	0.0210	0.0043	1.3E-06	0.2387	0.0293	**2.09E-18**
3	rs10589611	171,990 700	*FNDC3B*	TGA	T	1	0.2791	−0.0186	0.0027	**8.6E-12**	−0.0200	0.0037	**4.3E-08**	−0.2027	0.0246	**2.32E-18**
11	rs56118776	120,205 572	*TLDC5* ~*ARHGEF12*	GTATATA TATATATA	GTA	1	0.1830	0.0233	0.0032	**2.4E-13**	0.0210	0.0044	1.8E-06	0.2553	0.0295	**2.43E-18**
17	rs9913911	10,031 183	*GAS7*	G	A	0	0.4069	0.0176	0.0025	**4.0E-13**	0.0153	0.0033	5.6E-06	0.1906	0.0224	**1.30E-17**
6	rs2816294	1 538 781	*FOXC1*	A	G	1	0.4703	0.0155	0.0025	**2.5E-10**	0.0169	0.0033	3.1E-07	0.1751	0.0221	**4.55E-16**
12	rs671	112,241 766	*ALDH2*	G	A	0	0.1925	−0.0195	0.0031	**6.2E-10**	−0.0209	0.0042	8.7E-07	−0.2132	0.0279	**2.97E-15**
17	rs471064	4 898 533	*INCA1/KIF1C*	A	G	0	0.4956	0.0129	0.0025	1.7E-07	0.0155	0.0033	1.8E-06	0.1586	0.0221	**2.06E-12**
9	rs2422493	107,690 995	*ABCA1*	G`	A	0	0.3988	−0.0138	0.0025	**1.5E-08**	−0.0138	0.0034	4.0E-05	−0.1489	0.0225	**2.66E-12**
2	rs10181192	228,237 083	*TM4SF20*	T	C	1	0.5068	0.0121	0.0025	1.8E-06	0.0150	0.0033	4.2E-06	0.1362	0.0223	**5.39E-11**
9	rs8176747	136,131 315	*ABO*	C	G	1	0.1750	0.0205	0.0032	**4.9E-10**	0.0107	0.0043	1.3E-02	0.1794	0.0291	**7.67E-11**
21	rs57206396	47,548 398	*COL6A2*	G	GTGGCC TGGCCC	1	0.4321	−0.0104	0.0025	1.1E-05	−0.0152	0.0034	5.9E-06	−0.1394	0.0226	**5.28E-10**
8	rs4599779	108,299 577	*ANGPT1*	G	A	0	0.4470	−0.0111	0.0025	4.6E-06	−0.0136	0.0033	5.2E-05	−0.1288	0.0221	**1.20E-09**
5	rs202200816	132,129 659	*SEPT8*	T	A	1	0.2975	−0.0127	0.0027	3.0E-06	−0.0141	0.0036	1.1E-04	−0.1436	0.0242	**1.46E-09**
9	rs10780773	72,674 739	*MAMDC2*	C	T	1	0.4422	0.0109	0.0025	1.4E-05	0.0143	0.0033	2.0E-05	0.1266	0.0222	**1.81E-09**
8	rs185046339	6 373 094	*ANGPT2/MCPH1*	A	T	1	0.0126	−0.0549	0.0111	4.1E-07	−0.0435	0.0153	3.6E-03	−0.5235	0.1044	**6.13E-09**
11	rs111994526	47,558 817	*CELF1*	T	G	1	0.4167	0.0120	0.0026	1.9E-06	0.0109	0.0034	1.2E-03	0.1212	0.0231	**8.46E-09**
3	rs56054326	171,634 397	*TMEM212*	C	G	1	0.2990	−0.0134	0.0027	7.9E-07	−0.0105	0.0036	3.3E-03	−0.1390	0.0241	**1.03E-08**
2	rs163524	45,157 553	*LINC01833*	C	A	1	0.3128	0.0112	0.0026	2.2E-05	0.0135	0.0036	1.1E-04	0.1399	0.0239	**1.19E-08**
12	rs35186972	43,841 962	*ADAMTS20*	C	CA	1	0.1901	0.0144	0.0031	4.7E-06	0.0142	0.0042	6.9E-04	0.1547	0.0280	**1.22E-08**
17	rs111567274	58,893 676	*BCAS3*	G	A	1	0.1349	0.0160	0.0037	1.5E-05	0.0174	0.0049	3.7E-04	0.1780	0.0331	**2.25E-08**
22	rs62654724	46,375 833	*WNT7B*	C	T	1	0.3011	−0.0117	0.0027	7.3E-06	−0.0121	0.0036	8.7E-04	−0.1401	0.0244	**2.34E-08**

ANAPC1 = anaphase promoting complex subunit 1; *ARHGEF12* = Rho Guanine Nucleotide Exchange Factor 12; *ALDH2* = aldehyde dehydrogenase 2; *ABCA1* = ATP binding cassette subfamily A member 1; *ABO* = alpha 1-3-N-acetylgalactosaminyltransferase and alpha 1-3-galactosyltransferase; *ANGPT1* = angiopoietin 1; *ANGPT2* = angiopoietin 2; *ADAMTS20* = ADAM metallopeptidase with thrombospondin type 1 motif 20; *BCAS3* = BCAS3 microtubule associated cell migration factor; *CELF1* = CUGBP Elav-like family member 1; Chr = chromosome; *COL6A2* = collagen type VI alpha 2 chain; EA = effect allele; EA freq = effect allele frequency in tommo_38KJPN; *FNDC3B*: fibronectin type III domain-containing protein 3B; *FOXC1* = forkhead box C1; *GAS7* = growth arrest specific 7; *INCA1* = inhibitor of CDK, cyclin A1 interacting protein 1; IMP = imputed; *KIF1C* = kinesin family member 1C; *LINC01833* = long intergenic non-protein coding RNA 1833; MAMDC2 = MAM domain containing 2; MCPH1 = microcephalin 1; NA = not applicable; RA = reference allele; *SEPT8*: septin-8; SE = standard error; SNP = single nucleotide polymorphism; *TLDC5* = TLC domain containing 5; *TM4SF20* = transmembrane 4 L six family member 20; *TMEM212* = transmembrane protein 212; *WNT7B* = Wnt family member 7B.

Bold characters represent a statistically significant *P* value.

*The lead SNPs of each locus identified in the discovery analysis are presented.

†The position is based on NCBI human genome build 37.

‡The beta values for discovery and validation stages are derived from Box-Cox transformed statistics and are not expressed on the mmHg scale.

§The beta values were calculated by reanalyzing the data using plink2 to obtain values expressed on the mmHg scale.

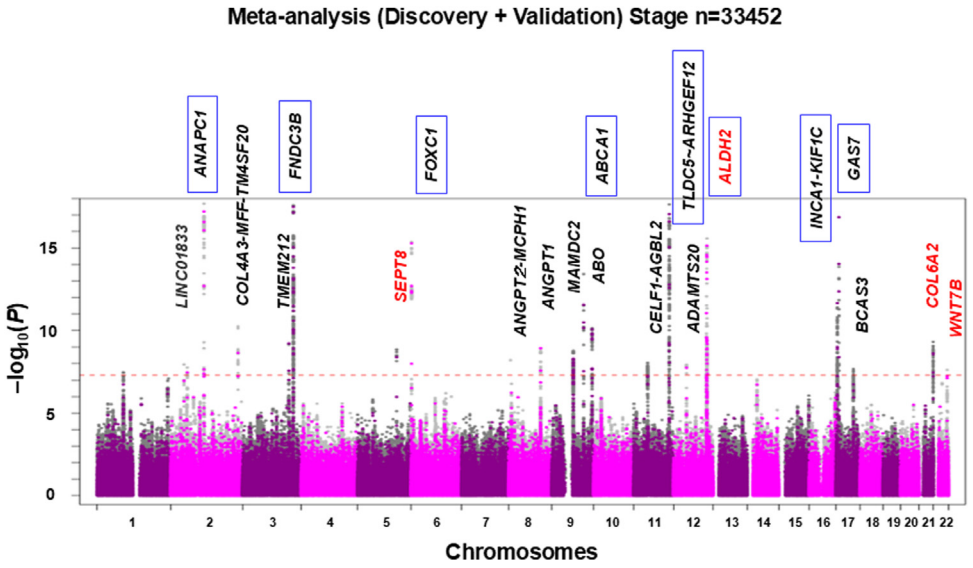


Figure 8. Genome-wide association study of intraocular pressure in the meta-analysis. The results of the GWAS for intraocular pressure in the meta-analysis are presented. Data for both directly genotyped and imputed SNPs are visualized in the Manhattan plot, with details consistent with those in Figure 7. Previously reported genes (loci) are labeled in black, while newly identified genes (loci) from this study are highlighted in red. Closed boxes indicate the top 8 SNPs. *ANAPC1* = anaphase promoting complex subunit 1; *ANGPT2* = angiopoietin 2; *ANGPT1* = angiopoietin 1; *ABO* = alpha 1-3-N-acetylgalactosaminyltransferase and alpha 1-3-galactosyltransferase; *ABCA1* = ATP binding cassette subfamily A member 1; *AGBLG2* = AGBL carboxypeptidase 2; *ADAMTS20* = ADAM metallopeptidase with thrombospondin type 1 motif 20; *ALDH2* = aldehyde dehydrogenase 2; BCAS3 = *BCAS3* microtubule associated cell migration factor; *COL4A3* = collagen type IV alpha 3 chain; *ARHGEF12* = Rho Guanine Nucleotide Exchange Factor 12; *CELF1* = CUGBP Elav-like family member 1; *COL6A2* = collagen type VI alpha 2 chain; *FNDC3B* = fibronectin type III domain-containing protein 3B; *FOXC1* = forkhead box C1; GWAS = genome-wide association study; *GAS7* = growth arrest specific 7; INCA1 = inhibitor of CDK, cyclin A1 interacting protein 1; *KIF1C* = kinesin family member 1C; *LINC01833* = long intergenic non-protein coding RNA 1833; *MFF* = mitochondrial fission factor; *MCPH1* = microcephalin 1; *MAMDC2* = MAM domain containing 2; SNP = single nucleotide polymorphism; *SEPT8* = septin-8; *TM4SF20* = transmembrane 4 L six family member 20; *TMEM212* = transmembrane protein 212; *TLDC5* = TLC domain containing 5; *WNT7B* = Wnt family member 7B.

the most significant association was identified in the *ANAPC1* locus (rs34739497; $P = 2.09 \times 10^{-18}$). Additionally, strong associations were observed in the *FNDC3B* locus (rs10589611; $P = 2.32 \times 10^{-18}$), the *TLDC5*∼*ARHGEF12* locus (rs56118776; $P = 2.43 \times 10^{-18}$), the *GAS7* locus (rs9913911; $P = 1.30 \times 10^{-17}$), and the Forkhead Box C1 (*FOXC1*) locus (rs2816294; $P = 4.55 \times 10^{-16}$), consistent with previous studies.[18,21]

Regional plots for the 8 top loci identified in the meta-analysis, based on direct genotyping and imputed genotype data, are presented in Figure S2 (available at www.ophthalmologyscience.org). Variations were observed in the number and distribution patterns of SNPs at each locus; however, all loci contained clusters of associated SNPs surrounding the most significantly associated SNP (lead SNP), represented by a purple diamond. These findings effectively validated the loci associated with IOP. Notably, the *GAS7*, *FOXC1*, and ATP binding cassette subfamily A member 1 (*ABCA1*) loci exhibited SNPs concentrated near the lead SNP, resembling a leptokurtic distribution. In contrast, the remaining 5 loci displayed a broader distribution of SNPs around their respective lead SNPs.

The Impact of Genetic Variants on IOP

Our analysis revealed that heterozygous variations in the top 8 SNPs identified in the meta-analysis stage, compared to the reference allele, induce statistically significant changes in IOP. The effects of these validated SNPs on IOP values are illustrated in Figure 9. Remarkably, all heterozygous variations demonstrated measurable effects, suggesting a dominant inheritance pattern for the associated phenotypes. Interestingly, the direction of the phenotypic effect varies depending on the genes in which the SNPs are located, allowing these genes to be categorized into 2 distinct groups. In the first group, heterozygous variations are associated with increased IOP, as observed in *ANAPC1, TLCD5*∼*ARHGEF12, GAS7, FOXC1,* and inhibitor of CDK, cyclin A1 interacting protein 1/kinesin family member 1C (*KIF1C*). Conversely, in the second group, heterozygous variations lead to decreased IOP, as seen in *FNDC3B, ALDH2,* and *ABCA1*.

We investigated whether the SNPs within the top 8 genes exhibited additive effects on IOP values as the number of variant alleles increased in homozygotes. Six genes demonstrated additive effects on IOP values, proportional to the number of variant alleles present. Three genes (*ANAPC1*, *ALDH2*, and *ABCA1*) did not show significant differences in IOP values between heterozygotes and homozygous variant alleles ($P > 0.05$). Overall, when comparing the additive and dominant models for the validated SNPs, the additive model appears to fit better (Table S2, available at www.ophthalmologyscience.org). For *ALDH2*, this observation aligns with previous findings demonstrating dominant inheritance patterns of *ALDH2*

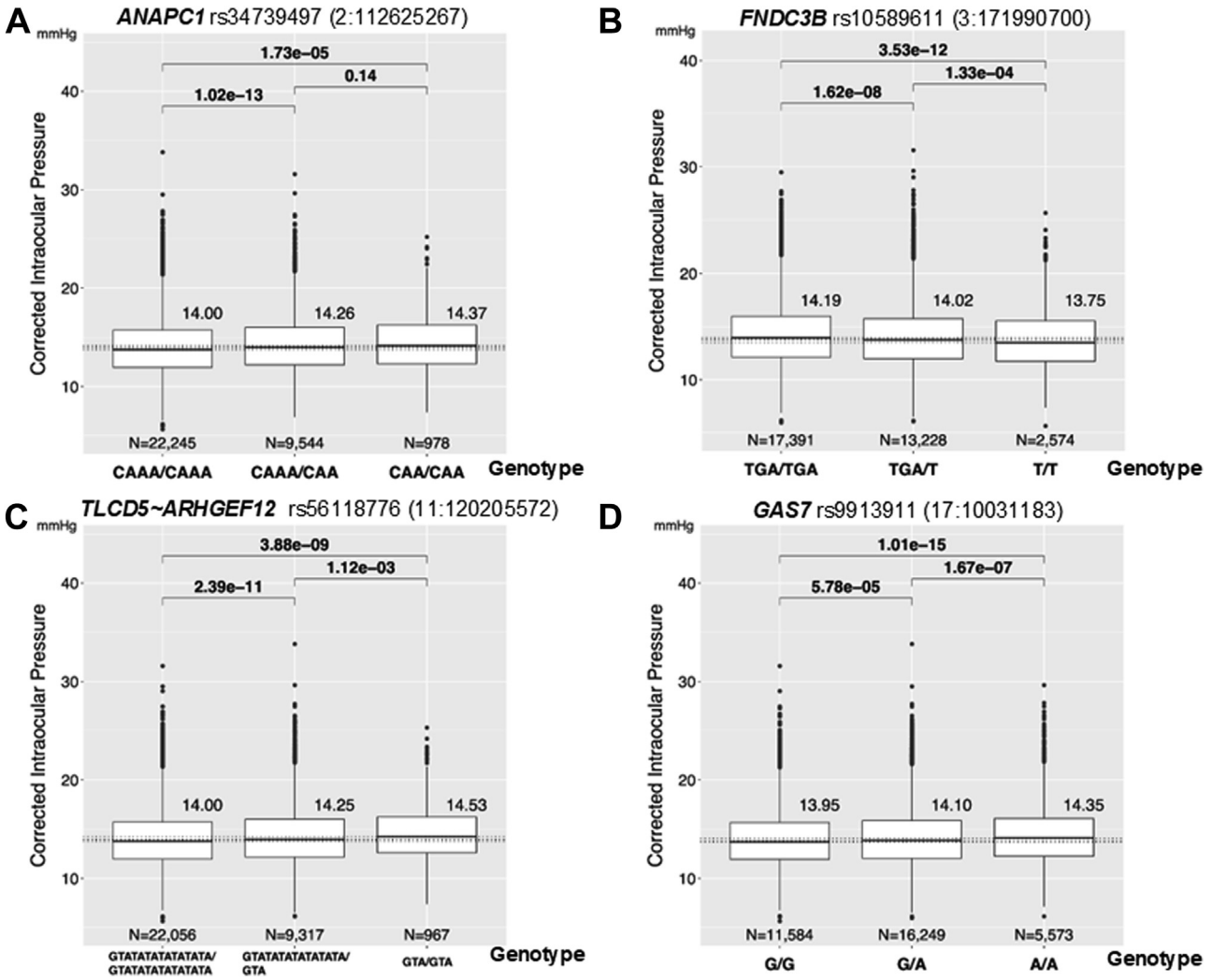


Figure 9. Effects of validated SNPs on the phenotypic values of IOP. The boxplot illustrates the effects of genotypes for the validated top 8 SNPs on IOP values. **A,** *ANAPC1* rs34739497, **B,** *FNDC3B* rs10589611, **C,** *TLCD5*∼*ARHGEF12* rs56118776, **D,** *GAS7* rs9913911, **E,** *FOXC1* rs2816294, **F,** *ALDH2* rs671, **G,** *INCA1/KIF1C* rs471064, and **H,** *ABCA1* rs2422493. All heterozygous variants exhibit measurable effects, categorized into 2 types: those associated with an increase in IOP (*ANAPC1*, *TLCD5*∼*ARHGEF12*, *GAS7*, *FOXC1*, and *INCA1/KIF1C*) and those associated with a decrease in IOP (*FNDC3B*, *ANAPC1*, *ALDH2*, and *ABCA1*). *ANAPC1* = anaphase promoting complex subunit 1; *FNDC3B* = fibronectin type III domain-containing protein 3B; *GAS7* = growth arrest specific 7; IOP = intraocular pressure; SNP = single nucleotide polymorphism.

variants in acetaldehyde metabolism.[49,50] However, the precise mechanistic basis by which SNPs in these genes influence IOP phenotypes remains unclear. Further investigation is required to determine whether these alterations in IOP are associated with the incidence or progression of glaucoma. These questions will be addressed in future longitudinal studies of the TMM cohort participants.

Discussion

To investigate the distribution of ocular biometric parameters and evaluate the association between IOP and SNPs, we analyzed IOP values in participants from Japanese population-based genome cohort studies. This analysis included 22 150 participants from the CommCohort and 11 302 participants from the BirThree Cohort of the ToMMo Eye Study. Unlike findings from the UK Biobank data,[43] our analysis demonstrated that IOP values decrease with advancing age. A GWAS was conducted using the CommCohort as the discovery stage and the BirThree Cohort as the replication stage. This analysis identified 573 SNPs with genome-wide significance in the discovery stage and 2 SNPs in the replication stage. Meta-analysis of these SNPs and their associated loci revealed 1601 SNPs distributed across 21 loci on 11 Chrs. Of these, 17 loci were previously reported as being associated with IOP or glaucoma, while 4 loci, *SEPT8, ALDH2, COL6A2,* and *WNT7B* were newly identified in this study. These findings highlight ethnic variations in age-related changes in IOP values and the genetic heterogeneity of SNPs associated with IOP.

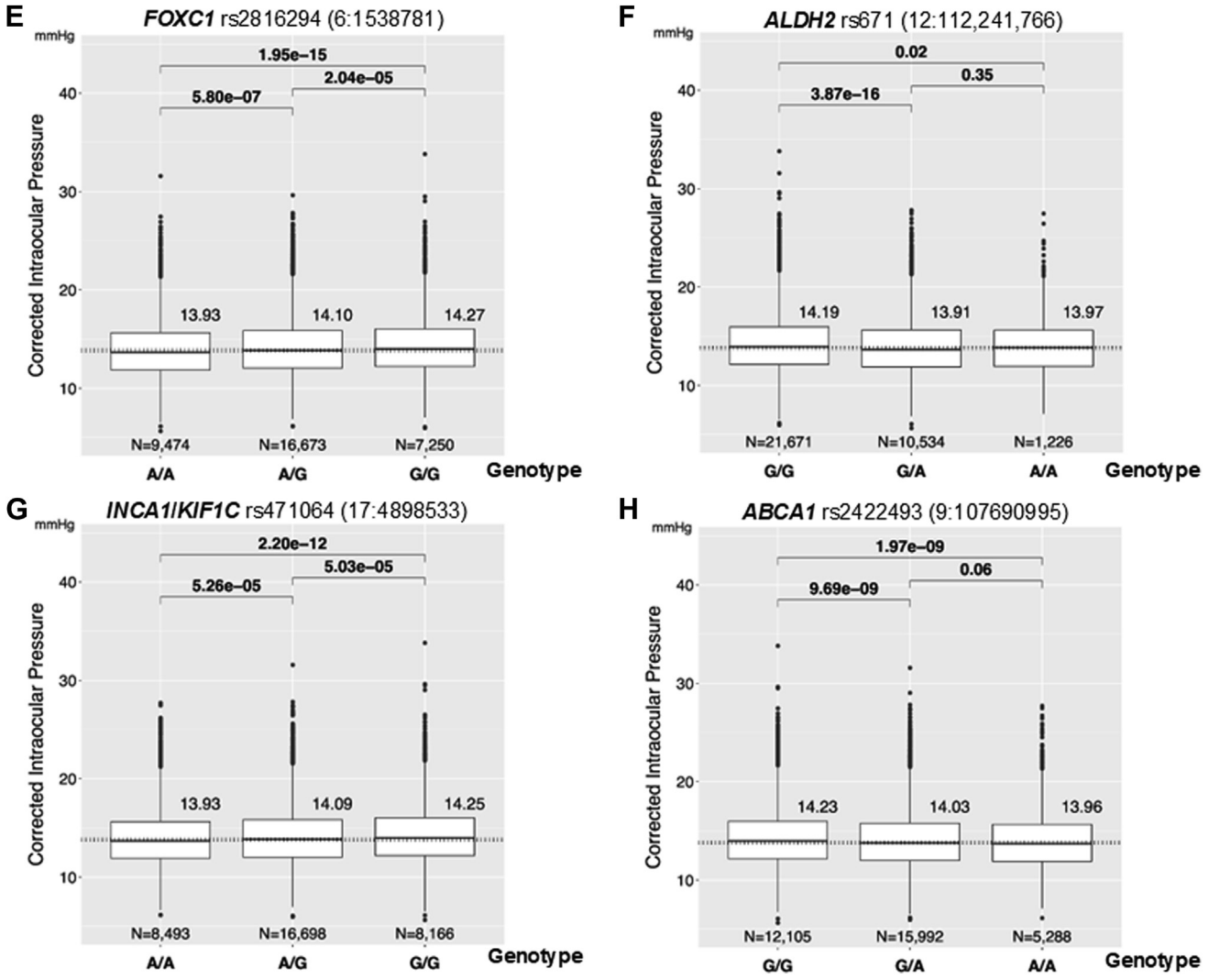


Figure 9. (*continued*).

These insights are expected to contribute to advancements in eye health care, particularly in the prevention and management of glaucoma.

Previous genetic studies have demonstrated a partial overlap in the genetic risk factors for POAG and IOP. For example, *TMCO1* and *GAS7* have been reported as associated with both IOP and POAG. The International Glaucoma Genetics Consortium identified novel IOP-associated loci near *FNDC3B*, caveolin (*CAV*)*1/2*, *ABCA1*, and others,[14] with common variants near *CAV1/2* also linked to POAG.[51] Similarly, *ANAPC1*,[18,21,46] *FOXC1*,[17,18,21,46] *ARHGEF12*,[45] *GAS7*,[16–18,21,46] ADAM metallopeptidase with thrombospondin type 1 motif 8,[16] calmodulin binding transcription activator 2 (*CAMTA2*),[17] have been reported as associated with both IOP and POAG. As for *FNDC3B* and *ANAPC1*, these genes are associated with CCT[52] and corneal biomechanical properties[53] that may affect IOP. Although we conducted the GWAS using IOP adjusted for CCT, the results should be interpreted with caution, as it may be primarily affecting the confounding effect on IOP. While the associations of these genes with IOP were replicated in our study, we did not observe a significant relationship between *CAV1/2* and *TMCO1* and IOP. Notably, this finding is consistent with previous Japanese studies, where *TMCO1* was not identified as a POAG-associated locus.[48] This suggests that these loci are almost monomorphic in Japanese or may not be associated with IOP and POAG in the Japanese population. These findings reinforce the hypothesis that the spectrum of IOP-associated and POAG-associated genes differs between Japanese and European populations, highlighting the importance of considering population-specific genetic variations in glaucoma research.

Among the 21 IOP-associated loci identified in this study, 4 loci: *SEPT8* (Chr5), *ALDH2* (Chr12), *COL6A2* (Chr21), and *WNT7B* (Chr22) are newly discovered.

Notably, *ALDH2* stands out as particularly intriguing. This gene belongs to the ALDH family, a key pathway for alcohol metabolism. Homozygotes for the minor allele of *ALDH2* rs671 exhibit lower IOP values compared to homozygotes for the major allele. Given that the minor allele homozygote is known to retain reduced enzyme activity for acetaldehyde metabolism, this finding suggests that diminished *ALDH2* activity might confer a protective effect against elevated IOP. Previous studies have reported that an association between habitual alcohol consumption is associated with increased IOP,[54] although no such relationship has been observed with glaucoma.[55] These findings imply a complex role for *ALDH2* in the regulation of IOP, warranting further investigation to elucidate the underlying mechanisms. Among the remaining 3 loci, *SEPT8* encodes a member of the septin family, a group of conserved GTP-binding cytoskeletal proteins involved in regulating cytoskeletal organization. Septin family proteins are expressed in various ocular tissues, including trabecular meshwork cells,[56] and may influence aqueous humor outflow in the anterior segment. *COL6A2* encodes 1 of the 3 alpha chains of type VI collagen. Previous GWAS have shown an association between *COL6A2* and CCT,[57] suggesting a potential physical influence of this gene on IOP values. Finally, *WNT7B* encodes secreted signaling proteins and has been identified as a susceptibility gene for AL and corneal curvature in the Japanese population.[26,58] These findings highlight a potential role for *WNT7B* in ocular biomechanics. This study reveals both genetic similarities and heterogeneity in SNPs associated with IOP variations. However, the precise mechanistic pathways by which variants in these genes influence IOP levels remain unclear and require further investigation.

Recent studies have demonstrated that polygenic risk scores for POAG are positively correlated with maximum IOP, disease severity, and the need for surgical intervention.[59] These findings suggest that the 21 IOP-associated genes identified in this study may hold clinical significance for the diagnosis and treatment of glaucoma. Interestingly, however, no variants in the CDKN2B and CDKN2A antisense cis and trans regulatory RNA 1 (*CDKN2B-AS1*) gene, reported as the gene most strongly associated with glaucoma risk in multiple GWAS across diverse populations,[15,19,22,48,60,61] were detected in this study. Similarly, variants in the *TMCO1* gene, which has also been linked to glaucoma risk, were not identified among the IOP-related variants in our analysis. *TMCO1* gene polymorphism is almost monomorphic in Japanese people, making it difficult to evaluate. We hypothesize that these results reflect distinct mechanisms underlying POAG pathogenesis. Specifically, POAG may involve 2 categories of variants: those associated with IOP regulation and those related to optic nerve vulnerability. Variants in CDKN2B-AS1 are likely to contribute to the latter category, highlighting the multifactorial nature of glaucoma susceptibility.

In this study, we observed a significant decrease in IOP with aging in both men and women during the discovery stage, a trend that was consistently replicated in the validation stage. However, the relationship between IOP and age remains complex and somewhat ambiguous. Several studies have reported no significant association between IOP and age,[62,63] while others have identified a negative correlation.[64,65] Another study described a reversed U-shaped trajectory, with IOP increasing up to the age of 60 to 64 years before declining thereafter.[66] In contrast, the large UK Biobank study, which included 112 285 valid measurements, demonstrated a linear increase in IOP with age.[43] This trend is further supported by 3 additional studies that reported positive associations between aging and elevated IOP.[67–69] Such findings may help explain the increasing prevalence of glaucoma with age in populations like the UK, where aging is considered a critical risk factor for POAG. Our present findings in a Japanese cohort, however, present a striking contrast. While glaucoma incidence also increases with age in Japan, our results suggest that certain cases of POAG in older Japanese individuals may not be associated with elevated IOP. Two key observations from this study support this hypothesis. First, the high prevalence of normal-tension glaucoma in the Japanese population has been well-documented.[44] Second, we recognized glaucoma-associated SNPs that were not linked to elevated IOP in this study. These findings highlight potential ethnic and genetic differences in the mechanisms underlying glaucoma pathogenesis, warranting further investigation.

The associations between IOP and background characteristics reveal that IOP increases with CCT, serum glucose, total cholesterol, platelet, lung function (percent-predicted forced expiratory volume in the first second), and systolic and diastolic blood pressure. Notably, diastolic blood pressure exhibited the strongest correlation with IOP in this study, aside from CCT. This contrasts with findings from the UK Biobank Eye Study, which identified systolic blood pressure as the most critical determinant of IOP.[43] Similarly, our observation aligns with previous reports emphasizing the stronger association of diastolic blood pressure with IOP.[70] However, the mechanistic underpinnings of these associations, as well as the ethnic differences, remain to be clarified. In contrast to the characteristics mentioned above, height was not found to be correlated with IOP in this study. This is consistent with findings from studies conducted on adult British[71] and Japanese populations,[64,65] which also reported no significant correlation between height and IOP. However, a study of Chinese Singaporeans aged 40 to 79 years indicated a positive correlation between these variables.[72] We propose that the relationship between height and IOP should be considered in the context of other morphological factors, such as body weight, to fully elucidate its potential role.

The TMM study uniquely includes a significant number of younger participants, aged 20 to 39 years, which distinguishes it from other general population cohort studies.[43,62,64–69] This recruitment strategy enables the analysis of younger generations, a demographic often underrepresented in similar research. Additionally, the study boasts a relatively large sample size compared to other eye-related cohort studies.[64–69] The response rate for participant recruitment has been approximately 65%, suggesting that the study population is representative of the broader

community.[24] This study employs a unified protocol to conduct IOP assessments and GWAS using an ethnic-specific DNA array developed specifically for the cohort.[27,28] This study also has limitations. Based on the questionnaire, participants with a history of glaucoma were excluded because they were unable to correct for IOP, as there were some cases where it was unclear whether they had undergone laser surgery, surgery, or only medication. The inclusion of participants with undiagnosed glaucoma may also have influenced the results. Furthermore, this analysis is based on single-visit measurements per participant, which could introduce variability. To address these limitations, the study has initiated follow-up examinations, with longitudinal analyses planned to provide a more comprehensive understanding of IOP and its genetic associations over time.

In conclusion, this study presents comprehensive ocular biometric data on IOP and genetic analysis findings from large voluntary cohorts within the adult Japanese population. These data serve as a valuable addition to existing IOP databases and enhance the catalog of IOP-related genetic loci. The genetic loci identified in this study provide deeper insights into the biological pathways influencing IOP and hold potential for improving glaucoma screening and predicting glaucoma risk in susceptible individuals.

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No animal subjects were used in this study.

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Abbreviations and Acronyms:
AL = axial length; ***ALDH2*** = aldehyde dehydrogenase 2; ***ANAPC1*** = anaphase promoting complex subunit 1; ***ARHGEF12*** = Rho Guanine Nucleotide Exchange Factor 12; ***ABCA1*** = ATP binding cassette subfamily A member 1; **BirThree** = Birth and Three-Generation; ***CAMTA2*** = calmodulin binding transcription activator 2; ***CAV*** = caveolin; **CCT** = central corneal thickness; ***CDKN2B-AS1*** = CDKN2B and CDKN2A antisense cis and trans regulatory RNA 1; **Chr** = chromosome; **CI** = confidence interval; **CommCohort** = Community-Based Cohort; ***COL6A2*** = collagen type VI alpha 2 chain; ***FNDC3B*** = fibronectin type III domain-containing protein 3B; ***FOXC1*** = Forkhead Box C1; ***GAS7*** = growth arrest specific 7; **GWAS** = genome-wide association study; **IOP** = intraocular pressure; **OMIM** = Online Mendelian Inheritance in Man; **POAG** = primary open-angle glaucoma; **SNP** = single nucleotide polymorphism; ***SEPT8*** = septin-8; **TMM** = Tohoku Medical Megabank; **ToMMo** = Tohoku Medical Megabank Organization; ***TLCD5*** = TLC domain containing 5; ***WNT7B*** = Wnt family member 7B.

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